

Examiner reports

Q1.

Most candidates lost marks on this question. Full marks were not common although they were seen from the more able candidates.

Not many candidates obtained full marks for part (a). Most candidates scored 2 out of 4 since they started correctly by differentiating $\ln t$ and integrating $(t - 1)$. Obvious errors were differentiating both terms or starting by trying to integrate $\ln t$. Several candidates managed to obtain the second accuracy (A) mark for simplification to $(t^2/2 - t) \ln t - \int (t/2 - 1) dt$ but lost the final accuracy mark by ending up with a final term of $-t$, not $+t$.

Part (b) was done well by most candidates, although poor manipulation often cost the loss of the final accuracy mark. Some candidates also confused the issue by trying to introduce terms in u and du .

Part (c) was not very well answered, with candidates failing to change the limits as a common error. Many candidates did not appreciate that they should be using their answer to part (a) and tried to start again, often obtaining different answers to those they had found in part (a). After correctly approaching the choice of u and dv/dt , most were then defeated by the required manipulation of the subsequent algebra.

Q2.

In part (a)(i), a few candidates ignored the request to use the factor theorem and scored no marks for using long division. It was necessary to make a statement that " $x + 1$ is a factor", after showing that $f(-1) = 0$, in order to score full marks.

Part (a)(ii) was not answered as well as similar questions in previous years. Perhaps the sketch lured some into trying to write down three factors without any further working, rather than using the intermediate step of showing that $p(x) = (x + 1)(x^2 - x - 6)$ before writing $p(x)$ as a product of three factors. Many who tried long division were flummoxed by there being no x^2 term.

In part (b)(i), those who had the correct linear factors in part (a)(ii) usually wrote down correctly that A had coordinates $(-2, 0)$, although some carelessly wrote the point as $(0, -2)$.

Many candidates simply found an indefinite integral in part (b)(ii) and then a definite integral in part (b)(iii). The two parts were generously treated holistically when candidates did this. The fractions once again caused problems to most candidates who are so used to having a calculator to do this work for them. It was very rare to see the correct answer of -32 for the definite integral.

In part (b)(iii), many lost out on an easy mark because they rolled their two sections into one: those who wrote "integral $= -32 = 32$ " gained full credit for part (b)(ii) but did not score the mark in part (b)(iii). It was necessary to give a positive value for the area of the region and to make this explicit. In anticipation of a lot of wrong answers in part (b)(ii), a follow through mark was awarded in part (b)(iii): for example, if a candidate's answer in part (b)(ii) was -20 and they concluded that the area was 20 in part (b)(iii), they scored the mark.

In part (b)(iv), most candidates differentiated correctly, but quite a few thought that $3(-1)^2 - 7$ was equal to -10 and thus obtained the wrong gradient of the curve.

In part (b)(v), a large number of candidates found the correct equation of the normal but some still confused tangents and normals and consequently thought that the gradient of the normal was equal to -4 . It was quite common for weaker candidates to either negate their gradient or take the reciprocal but to fail to do both.

Q3.

Generally, most candidates scored relatively high marks for this long structured calculus question. Parts (a)(i), (ii) and (iii) were answered very well, although some used the gradient of the normal in part (a)(iii) as -2 and so in part (a)(iv) obtained a positive value for the y -coordinate of Q . A glance at the diagram should have indicated that such a value was incorrect.

Part (a)(v), which required candidates to find the x -coordinate of the maximum point, M , was generally not answered with any confidence. To score any marks for this part, candidates were required to go beyond just equating their answer for part (a)(i) to zero. Many candidates again incorrectly believed that for a maximum point the second derivative was zero. Candidates generally found the correct answer to the indefinite integral in part (b)(i), but a significant minority forgot to add the area of the triangle to their 'area under the curve' in part (b)(ii) of the question.

Q4.

Part (a)(i) was well answered.

Candidates who saw the connection with part (a)(i), usually made a very good attempt at part (a)(ii), often with complete success. Limits did cause a problem where candidates had not bracketed $2x^2 - 8x + 3$, and some candidates did not leave the answer in the required

form but left their answer as $\frac{1}{4} \ln 9$.

Part (b) was well answered by candidates who were able to write the integral in terms of u successfully. Many candidates lost a mark through omission of either du or c . A disturbing

number of candidates thought that $\frac{1}{3} \times \frac{1}{3} = \frac{1}{6}$. Some clearly able candidates lost the last mark by spoiling an otherwise acceptable answer.

Q5.

Part (a) was usually answered well and several correct equations were seen. However

errors such as $\frac{-15}{-3} = -5$ sometimes occurred, and weaker candidates confused coordinates and wrote equations such as $y - 15 = m(x + 2)$. Some attempted to differentiate to find the gradient instead of using the coordinates of the points.

Although the integration in part (b)(i) was largely successful, the evaluation of $F(1) - F(-2)$ defeated most. Arithmetic errors, usually caused by minus signs and fractions, abounded.

Some assumed that they had answered part (b)(ii) by substituting the limits after integration. Of those who realised the need to subtract the area of a triangle, many assumed the triangle base to be 4 rather than 3. Another error seen was the integration of the curve from -2 to $+2$ followed by subtracting their answer to part (b)(i). Some successfully found the area under the line by integration, but that was not the expected method.

Q6.

Although many candidates applied a correct method to find the expansion of $\left(1 + \frac{4}{x^2}\right)^3$, accuracy marks were sometimes lost because brackets were missing and the factor 4 was not squared when finding the value of q . Although a significant minority of candidates failed to use the expansion in part (a), despite the word 'Hence', most other candidates obtained the method mark in part (b), but it was surprising to find candidates having more difficulty integrating '1' than integrating the terms in x . Many candidates scored the method mark in part (b)(ii) but arithmetical errors were common in the subsequent evaluation.

Q7.

Part (a) of this question was reasonably answered with most candidates obtaining the method mark for the product rule. Some candidates then lost the accuracy mark through incorrect evaluation of the constant associated with the derivative of $\cos 2x$; 2 and $\frac{1}{2}$ were frequently seen.

Answers to part (b)(i), although frequently correct, were often badly set out with the function being equated to zero or x being changed to α at various points in the solution. Division of $\cos 2x$ by $\cos 2x$ often resulted in zero before going on to fudge the correct answer. Part (b)(ii) was usually well answered with correct evaluations of $f(0.4)$ and $f(0.5)$. Some candidates then lost marks by just saying α was 'between these two values' without stipulating 0.4 and 0.5. Part (b)(iii) was usually well done but there were many cases when the $\tan$ became removed from its $2x$ and $\frac{1}{2} x \tan^{-1} = x$ was often seen. Part (b)(iv) was generally well answered but marks were lost from the use of degrees and for not writing the answer to the required degree of accuracy.

Many fully correct answers and many answers which only lost the final accuracy mark were seen in part (c). Other answers which had the wrong coefficient associated with the $\sin 2x$ often got the method marks. There were candidates who started with $u = x$ and $v =$

$\cos 2x$ or started with $\frac{dv}{dx} = x$ and obtained a more difficult integral.

Q8.

Part (a)(i) Some candidates ignored the request to state the coordinates of B even though they were using the height of the triangle as 5. The negative x -coordinate of A caused quite a few to feel that the triangle had a negative area. Far too many when finding

$\frac{1}{2} \times 1 \times 5$ wrote the answer as 3.

Part (a)(ii) Practically every candidate found the correct integral although some made errors when cancelling fractions.

Part (a)(iii) It was necessary here to have the lower limit as -1 and the upper limit as 0 . Many reversed the order and by some trickery arrived at a positive value. This was penalised and so very few, even though many had an answer of 1 for the area, scored full marks for this part of the question.

Part (b) Most candidates differentiated correctly but, because of poor understanding of negative signs, many wrong values of -13 were seen for the gradient. There is obviously confusion for many between tangents and normals and several thought the gradient of the

tangent was $\frac{-1}{17}$.

Q9.

This calculus question again proved to be a good source of marks for many candidates.

Part (a)(i) was answered very well but in part (a)(ii) a significant minority used $\frac{d^2x}{dx^2} = 0$ instead of $\frac{dy}{dx} = 0$ dx^2 . In part (a)(iii), although most candidates showed that they understood the method to find the gradient of the tangent, a significant proportion failed to find and use the gradient of the normal. Many candidates showed that they had a good understanding of integration but, surprisingly, errors in integrating $(x + 1)$ were almost as common as errors in integrating $\frac{4}{x^2}$. Examiners expected to see some simplification of the algebraic term $\frac{+4x^{-1}}{-1}$. In the final part, the evaluation of $F(4) - F(1)$ as 8.5 was the most common error.

Q10.

Part (a)(i) was well answered, with the main error being $\int -e^x dx = -2e^{2x}$. In part (a)(ii) the answer was given. Frequently candidates tried to 'fudge' their solutions to arrive at the correct result, with common error being $4 \ln 2 - 2 - \frac{1}{2} = 4 \ln 2 - \frac{3}{2}$, the given answer.

Part (b)(i) was again well answered by the majority of candidate, although in part (b)(ii) the step from $2x = \ln 4$ to obtain $x = \ln 2$ was often unconvincing. Some candidates answered part (c) very well and gained full credit. The main error was using 8 or -8 as the gradient of the normal, however many candidates earned at least partial marks.

Although correct answers were seen in part (d), they were not common. Many candidates earned the method mark by the substitution of $x = 0$ and some went on to find the area of

the triangle. The final mark was often lost through a failure to realise that the area was positive.

Q11.

Part (a)(i) Most candidates realised the need to find the value of $f(x)$ when $x = 1$. However, it was also necessary, after showing that $f(1) = 0$, to write a statement that the zero value implied that $x - 1$ was a factor.

Part (a)(ii) Those who used inspection were the most successful here. Methods involving long division or equating coefficients usually contained algebraic errors.

Part (a)(iii) This section seemed unclear to some candidates. Many tried to find the discriminant but used the coefficients of the cubic equation. Many who used the quadratic thought that in order to have one real root the discriminant had to be zero, no doubt thinking the question was asking about equal roots. Some correctly stated that 1 was the only real root but many were obviously confused by the terms “factor” and “root” and stated that “ $x - 1$ was a root”.

Part (b)(i) Most candidates were well versed in integration and earned full marks here.

Part (b)(ii) The correct limits were usually used, although many sign/arithmetic slips occurred after substitution of the numbers 1 and 2. Very few candidates realised the need to find the area of a triangle as well and so failed to subtract the value of the integral from the area of the triangle in order to find the area of the shaded region.

Q12.

Most candidates were able to correctly simplify the given expressions although $x^{\frac{9}{4}}$ was a

common wrong answer in part (a)(iii) for the simplification of $\left(x^{\frac{3}{2}}\right)^2$. The majority of candidates displayed a good understanding of integration but a significant proportion of

the entry left their answer to part (b)(i) in the form $\frac{3}{2}x^{\frac{3}{2}}$. The final mark of three required the answer to be simplified further and the constant of integration present. The majority of candidates showed sufficient working in part (b)(ii) to show that they had used their answer to part (b)(i) and so gained the method mark, but a significant minority failed to evaluate $2 \times 9^{\frac{3}{2}}$ correctly.

Q13.

Many candidates answered part (a) correctly and scored full marks but there were a significant number who gave a denominator of 10 in the second term of the answer after an otherwise correct approach. Some candidates tried to integrate x and differentiate e^{5x} which thereafter gave them no further credit.

In part (b)(i), most candidates obtained marks for finding $\frac{dz}{dx}$ but their attempts to obtain the

given answer were often flawed. Part (b)(ii) was not done particularly well; many split the denominator, and did not see that it required the use of $\ln$ at all.

Q14.

Part (a) was very well answered by the majority of candidates although errors of $\tan^2 x$ and $\ln(\sec x)$ were seen.

In part (b), the first step, the use of the quotient rule, was reasonably well done although the quotient rule was not always quoted correctly. Errors occurred where candidates had wrong signs for derivatives of $\sin x$ and $\cos x$ or where the functions in the numerator were reversed. Many candidates went on and used appropriate trigonometric identities to obtain the given answer.

Part (c) was not very well answered by most of the candidates. Many candidates obtained the correct expansion of $(\tan x + \cot x)^2$ but were then unable to handle the term $2 \tan x \cot x$. There were numerous flawed attempts using expressions such as $(\tan^2 x + 1) + (\cot^2 x - 1)$ equals the answer given.

In part (d), many candidates obtained the first method mark but went no further. Those who correctly integrated the expression often lost the final accuracy mark by working in degrees, with 57° being common.

Q15.

(a) Not everyone recognised that the height of the rectangle was the value of y when $x = -1$ or $x = 4$. Some who did made numerical errors. Even having found the height 4, some obtained the wrong area by taking AB as 4 or 2 (sometimes using Pythagoras' Theorem) or by finding the perimeter instead.

(b) (i) The integral was generally correct, though sometimes incorrect simplification occurred subsequently. A few integrated x^3 to $\frac{1}{4}x^4$ and a surprising number misread the integrand as $3x^2 - x^2$. There were also candidates who confused integration with differentiation or whose process was a hybrid of the two.

(ii) Almost everyone recognised that they should firstly evaluate the integral from -1 to 2 , but most stopped there, instead of going on to subtract the value of the integral from the area of the rectangle. There were a lot of sign errors in the work with some adding instead of subtracting or putting the two parts the wrong way round. A few wrongly substituted in the original function. Those who chose to work with the differences of two integrals seldom completed it correctly.

(c) (i) Differentiation was done well on the whole.

(ii) Many substituted $x = 1$ into the derivative to find the gradient of the tangent

and went no further. Most did not find the y coordinate of the point and so made no attempt at the equation of the tangent. A few non-linear equations were seen with a 'gradient' of $6x - 3x^2$.

- (iii) Very few candidates completed this part. Many made no attempt, and those who did tended to test a few values of x or to find the second derivative, which was of no value.

Only the strongest candidates realised that $\frac{dy}{dx} < 0$ was the condition for y to be decreasing and that, after a couple of lines of algebra, the given inequality could be obtained.

- (d) Although most made an attempt at the quadratic inequality, few obtained both parts of the solution. It was imperative that candidates wrote $x > 2$, $x < 0$ and not $0 > x > 2$ as many incorrectly stated.

It was disappointing to see how many candidates at this level could not solve the equation $x^2 - 2x = 0$, obtaining values such as -2 , $\sqrt{2}$, $1+\sqrt{2}$. Using the formula or completing the square sometimes led to $1 \pm \sqrt{1}$, which many candidates failed to simplify.

Q16.

This final question was a good source of marks for many candidates. Even the weaker candidates were able to give correct answers for parts (a) and (c) which tested standard C2 work on differentiation and integration. A significant minority of candidates started part (b)(i) by equating dy/dx to zero instead of substituting $x = 0$ to find the value of the derivative. Those who were able to answer the first two parts of part (b) correctly generally went on to find the correct coordinates of P . A significant number of candidates left their final answer as 24.3 without considering the area of the triangle OAP and carrying out the subtraction. For those candidates who attempted to find the area of the triangle some applied very lengthy methods which involved finding the length of each side of the triangle

then using the cosine rule to find angle OPA then using the formula $\frac{1}{2} ab \sin C$ or involved the integration of the equations of the two

lines. To their credit a number of such candidates did avoid premature approximation to reach the correct value for the area of the triangle which could have been obtained more

directly by using $\frac{1}{2} OA \times |y_P|$.

Q17.

- (a) (i) This was reasonably well done, although some candidates integrated.
- (ii) Many candidates failed to appreciate the significance of part (a)(i) and obtained complicated incorrect expressions. Those candidates who did appreciate the relevance of part (i) either obtained the correct answer or $2\ln |x^4 + 2x|$.

- (b) (i) This was fairly well answered. However, many candidates lost marks because they did not realise that in any substitution question, apart from replacing functions of x with functions of u , they have to find and use $\frac{du}{dx}$.
- (ii) The integration was very well answered by the majority of candidates with many accurately completing the question. Problems did arise through the use of incorrect limits. Many candidates converted their integration answers in u to terms in x and then used the original limits. This method was equally successful.

Q18.

- (a) Reasonable attempts were made by most candidates. The main mistake was finding the derivative of x^{-2} which was often seen as $-2x^{-1}$. This lost marks since candidates then 'found' the answer given by incorrect means.
- (b) There were many fully correct answers but many candidates chose incorrect functions for u and $\frac{dv}{dx}$, with the obvious implications.
- (c) (i) This was reasonably well answered, the main error seen being $1 - 21nx = x^3$.
- (ii) The majority of candidates obtained the method mark following through from their answer to part (b). Few candidates completed this question correctly.

Q19.

Part (a) was usually correct. Some candidates just wrote the result down whilst others derived it from $\cos^2 x - \sin^2 x$. The common error was to have $2 \cos^2 x + 1$, or the 2 missing. There were many fully correct responses to part (b) and nearly all candidates made some attempt to make use of the double angle in their integration. Accuracy sometimes slipped with the $1/2$ attaching to the $\cos 2x$ term only. Virtually all candidates gave evidence of correct use of limits, but those who didn't obtain the requested form of $\frac{\pi}{\alpha}$, showed no attempt to find their error.

Q20.

Part (a)(i) Most candidates differentiated correctly. However a few made a slip or misread one of the terms.

Part (a)(ii) Although most managed to substitute 2 into their derivative some made numerical errors and some used y or the second derivative. Most who realised that they should equate their derivative to zero (or at least showed their intention though never inserting the $= 0$) then tried to factorise or use the formula (though it was clear that some did not recognise the quadratic equation as such). It was disappointing that 14 the bracket $(3x - 14)$ often produced the solution $x = 14$ instead of $14/3$, and the solution of $x = 2$ did not always appear.

Part (b)(i) The integration was also completed correctly by most candidates, although the $28x$ was occasionally wrong and some 'hybrid' processes led to terms such as $\frac{20x^3}{3}$

In part (b)(ii) almost everyone attempted to substitute 3 into their integral but their problems with the ensuing fractions often took pages to resolve, and although most ended 'magically' with the required answer there were many errors en route. A few substituted into the original expression for y instead of the integrated expression.

Part (b)(iii) This part was quite well done although again there were errors in evaluating

both $\frac{1}{2} \times 21 \times 3$ and $56 \frac{1}{4} - 31 \frac{1}{2}$. Some candidates confused length with area and merely used Pythagoras's Theorem to find the length of the hypotenuse of the triangle. Those who chose to integrate the equation of the straight line were sometimes successful but many made arithmetic errors.

Q21.

Many candidates lost marks on this question from careless work or failure to write answers to the correct degree of accuracy.

Part (a) Although many candidates obtained the correct method mark

$\frac{dy}{dx} = x \times \frac{1}{x} + \ln x$, a significant number who reached this stage were unable to cope with $x \times \frac{1}{x}$ and equated it to 0, giving an answer of $\ln x$.

Part (b) Many candidates failed to realise the connection with the previous part. A significant number of candidates just wrote down the correct answer with no method indicated.

Part (c) Many candidates scored the method mark; some for the correct substitution and many for following through their answer to (b). Some candidates lost the accuracy mark for not writing the answer in exact form but going straight from $(5\ln 5 - 5) - (1\ln 1 - 1)$ to 4.047. Surprisingly, a number of candidates could not evaluate $-5 - (-1)$ correctly.

Q22.

- (a) (i) Some were confused by the word "factor" and gave answers as 1 and -2 .
- (ii) Although there were many correct answers given for the factorisation, some candidates appeared to be guessing the factors of $f(x)$, making no use of the information given in part (a)(ii). Despite getting a non-zero remainder for part (a)(i), several candidates gave $(x + 1)$ as a factor.
- (b) Most candidates found the coordinates of A , and those with the correct factor $(x - 4)$ of $f(x)$ were usually able to find the correct coordinates of B .
- (c) It was pleasing to see most candidates able to integrate correctly, although quite a few failed to integrate 8 to give $8x$. Some candidates did not evaluate the limits in the correct order, but there were many correct values seen for the area of the

shaded region.

Q23.

Parts (a) and (b) were answered very well and although the powers in answers to part (c) were nearly always correct, the coefficients were sometimes the reciprocal of the powers in part (b). Part (d) caused significantly more problems to the candidates. Although the method of solving a definite integral was well understood, many candidates resorted to using decimal approximations and, in general, candidates displayed a weakness in converting fractional powers of numbers back into surds.

Q24.

This question was answered very well by many candidates, and evidently the basic processes of differentiation and integration had been well rehearsed.

Part (a)(i) Most candidates substituted $x = 3$ into the given cubic equation and found $p(x)$ to be zero. Some, however, lost a mark as they did not indicate that $p(3) = 0$ implied that $x - 3$ was a factor.

Part (a)(ii) The coordinates (3, 0) were often seen but some forgot to give the y coordinate

Part (b)(i) Almost all candidates differentiated correctly, which was pleasing.

Part (b)(ii) Most candidates realised the need to equate dy/dx to zero. However not all were able to solve the resulting equation. Some of those who did solve the quadratic offered both solutions, 1 and $7/3$. Many found the second derivative in order to ascertain which point was the minimum point M . This was unnecessary as the coordinates of A (1, 0) were clear from the diagram.

Part (c) The second derivative was almost always found correctly followed by the substitution of $x = 1$.

Part (d)(i) The integration was carried out very well. Occasionally $\frac{3x^4}{4}$ was seen but this was rare.

Part (d)(ii) Most candidates used the correct limits of 0 and 1. However many arithmetic errors occurred.

When trying to combine the fractions so that the correct value of the integral $-\frac{11}{12}$ was not seen very often. Of those who did integrate correctly, many still lost the last mark for not stating that the area of the shaded region was $11/12$ and the negative value of the integral arose because the region was below the x -axis. It was not acceptable to simply

write Area = $-\frac{11}{12} = \frac{11}{12}$.

Q25.

This question proved to be a good source of marks for many candidates. In part (a)(i)

most candidates quoted the correct value for the power p . Although many candidates produced correct answers for the integration, all the usual errors, as illustrated by

$$\frac{3}{2}x^{1.5}, x^{1.5}, \frac{x^{-0.5}}{-0.5}, \text{ and } \frac{\sqrt{x}^2}{2},$$

were seen. Most candidates stated and used the correct limits in part (a)(iii) but evaluation was sometimes poor. Many of the candidates found a correct equation for the tangent but it was surprising to find a significant minority of these candidates either making no attempt to find the area of triangle AOB or giving the area as a negative value. For full marks in part (c) the examiners were expecting to see the word 'translation'. Some candidates wrote 'tr' but this was not considered to be clear enough to distinguish between 'transformation' and 'translation'. Common wrong answers involved 'stretch' and vectors in the wrong direction. In part (d) there was a significant minority of

candidates starting their solution with the incorrect statement $\sqrt{x-1} = \sqrt{x} - \sqrt{1} = \sqrt{x} - 1$ but in general the trapezium rule was well understood and many candidates were able to give the final answer to the correct required degree of accuracy in this numerical integration question.

Q26.

This question discriminated well between candidates. Part (a) showed an alarming lack of the correct use of mathematical language. Candidates had to use the words translation and stretch. Centres must stress this to all their students. There were many correct solutions to part (b) although some candidates missed the final mark by giving their answer as a decimal rather than the exact value that was required. Many candidates scored some marks on the final part but there were few fully correct solutions.

Q27.

Part (a) Most candidates successfully wrote down the expression for $\sin 2x$.

Part (b)(i) This caused difficulties for some candidates. Fairly common answers were $2\cos 2x = 2\cos^2 x - 2\sin^2 x$ or $\cos 2x = \cos^2 x + \sin x$.

Part (b)(ii) Most candidates made a reasonable start at proving the identity with many more able to handle the given $x + 2x$ than could consider $x + x$ for themselves. Candidates' responses varied considerably from clear concise derivations, to dead ends and restarts, to long meandering routes; some expanding $\cos 2x$ and then bringing it back in. Most candidates knew they had to eliminate $\sin x$ at some point, but this was not always done convincingly.

Part (c) Those candidates who saw the connection to part (b)(ii) were usually successful in their integration, although there were some sign errors and also errors in handling the 4. Some candidates wrote $\cos x^3$ as $(1 - \sin^2 x) \cos x$ and went on to integrate correctly. Attempts by parts, and even by substitution, went awry. Many candidates opted for the somewhat hopeful $\sin^4 x$, or similar, as their integral but could get some credit for correct use of the given limits.